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Role of the OSI Model in Software-Defined Networking (SDN)

CSA0711 — COMPUTER NETWORKS

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Introduction to OSI Model & SDN

- The OSI Model is a 7-layer reference framework that standardizes how data moves from one device to another across a network.
- Traditional networks bundle the control logic (decision-making) and data logic (packet forwarding) together inside every switch and router.
- The OSI layers still describe how data physically and logically travels; SDN changes only who makes the forwarding decisions and how.

EXAMPLE

A traditional switch decides where to send a packet using logic built into its own hardware (Layer 2/3).

In an SDN network, that switch simply asks a central controller — which sits logically above Layer 3 — “where should this packet go?” and forwards accordingly.

The OSI Model — Seven Layers

Layer 1 (bottom) moves bits; Layer 7 (top) serves the application

7	**Application**	End-user services — HTTP, DNS, network apps
6	**Presentation**	Data formatting, encryption, compression
5	**Session**	Establishes and manages communication sessions
4	**Transport**	Reliable end-to-end delivery — TCP, UDP
3	**Network**	Logical addressing and routing — IP, routers
2	**Data Link**	Framing and MAC addressing — switches
1	**Physical**	Raw bit transmission over cable, radio, fiber

KEY POINTS

- Layers 5–7 (upper) handle application-facing logic
- Layers 1–4 (lower) handle transmission and addressing
- SDN's control plane logically sits above Layer 3
- SDN's data plane operates at Layers 1–4
- Each layer only talks to the one directly above or below it

SDN Architecture: The Three Planes

Each plane corresponds to a different band of the OSI stack

Application Plane

OSI: Above Layer 7

- Hosts network apps: firewalls, load balancers, analytics
- Talks to the controller via Northbound REST APIs
- Defines what the network should achieve, not how

Control Plane

OSI: Logically above Layer 3

- The SDN controller — the network's centralized “brain”
- Computes routing/forwarding decisions for the whole network
- Talks to switches via Southbound protocols like OpenFlow

Data Plane

OSI: Layers 1–4

- Physical and virtual switches that forward packets
- Executes flow rules pushed down by the controller
- Has no independent decision-making of its own

Conclusion

KEY TAKEAWAYS

- The OSI Model still describes how data physically travels across any network, SDN included
- SDN separates the Control Plane (decision-making) from the Data Plane (forwarding)
- The Control Plane sits logically above OSI Layer 3; the Data Plane spans Layers 1–4
- Northbound and Southbound APIs (like OpenFlow) connect the planes across those layers

THANK YOU


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